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In Vitro Models to Study Human Lung Development, Disease and Homeostasis

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The main function of the lung is to support gas exchange, and defects in lung development or diseases affecting the structure and function of the lung can have fatal consequences. Most of what we currently understand about human lung development and disease has come from animal models. However, animal models are not always fully able to recapitulate human lung development and disease, highlighting an area where in vitro models of the human lung can compliment animal models to further understanding of critical developmental and pathological mechanisms. This review will discuss current advances in generating in vitro human lung models using primary human tissue, cell lines, and human pluripotent stem cell derived lung tissue, and will discuss crucial next steps in the field.

Lungs have evolved to serve the essential function of extracting oxygen from the air for use in aerobic metabolism and to remove the gaseous waste of this process from the body. Large, multicellular animals like humans require an enormous quantity of oxygen to maintain baseline energy levels needed for survival (2, 13). To meet these energy demands, lungs have evolved to maximize the surface area available for gas exchange by forming a complex network of tube-like epithelial branches known as the conducting airway, which consists of the trachea, bronchi, and bronchioles (FIGURE 1). The tubes in this branched network get progressively smaller until they terminate with thin distal air sacs, called alveoli, which are closely associated with the capillary network to allow diffusion of oxygen into the bloodstream and removal of carbon dioxide (FIGURE 1). When lung function is compromised, whether due to developmental defects or disease, the consequences can be severe and are often fatal.

Animal models have been instrumental in our understanding of lung development and disease (26, 99, 111). However, given many differences between animal and human lung physiology, there is an unmet need for human lung model systems that can complement animal models to improve our understanding of human lung physiology. Indeed, genetic gain- and loss-of-function studies in mice have improved our understanding of signaling networks in the developing lung and have shed significant light on physiological processes such as branching morphogenesis, which generates the arborized tree-like network of the lung (26, 106, 111). Similarly, rodent lung injury models can reproduce some aspects of complex human diseases such as lung fibrosis and emphysema (109, 142), and have helped illuminate multiple disease mechanisms. However, significant differences in lung physiology between mice and humans exist, which complicates inferring how animal studies can be applied to human development and disease (4). For example, mouse lungs develop quickly and do not begin forming alveoli until after birth, whereas human lungs undergo many additional rounds of branching before beginning alveolarization (130). Human lungs also have anatomical differences in the localization of stem and progenitor cells compared with rodent lungs. Thus it is not surprising that animal models are not able to recapitulate the full spectrum of human pathophysiology (97, 114), which is further highlighted by the fact that up to 80% of drugs that pass preclinical animal tests in rodents fail to effectively treat human disease during clinical trials (125).

Human model systems that reliably and accurately recapitulate the complexities of human biology in vitro should help improve our understanding of and ability to treat human lung disease. In vitro models of the human lung are also likely to provide powerful platforms for large-scale screens, molecular level analysis of cell-cell interactions, and the potential to study or treat lung disorders with personalized medicine.

In this review, we will discuss the current state-of-the-art for systems to model the human lung in vitro. Since human systems are being developed to mimic human lung development and homeostasis/disease in the adult, this review will briefly discuss lung development, adult homeostasis, and injury repair followed by a discussion of current in vitro models of the human lung using immortalized

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cell lines, primary patient-derived tissue and pluripotent stem-cell-derived human lung tissue. We will further discuss how these systems can be implemented to better understand human pathology, evaluate benefits and drawbacks of the current models, and explore critical next steps for the field.

Human and Murine Lung Development

Proper lung formation during development is critical to survival after birth. Lung organogenesis begins around *embryonic day* (E) *9* in the mouse and around *week 3* during human development as the primitive lungs bud from the foregut endoderm (39, 41, 111). As the lungs develop, they undergo branching morphogenesis, which generates a stereotyped network of epithelial tubes surrounded by mesenchymal tissue (106). During this time, the epithelial tubes are patterned along the proximal-distal axis of the lung (FIGURE 1) (111). The proximal airway is made up of proximal progenitor cells, which eventually give rise to mature cell types of the bronchi, whereas the distal airway is made up of distal progenitor cells, which maintain their proliferative progenitor state at the tips of branching buds until the branching program is completed. Distal airways ultimately give rise to alveoli (27, 132). Multiple pathways and transcription factors are critical for establishing proximal-distal patterning, controlling progenitor states and regulating branching morphogenesis. Regulation of branching morphogenesis has been reviewed extensively in numerous excellent reviews (26, 41, 76, 111, 134, 136). After the branching program, which lasts from E12.5 to E17.5 in mice and from *weeks 5* to *26* in humans, the distal airways undergo sacculation (40, 130, 136). During this process, distal epithelial progenitors begin to differentiate into the specialized cell types of the alveoli and undergo morphological changes to take on a sac-like structure. Full maturation of the alveolar sacs occurs from birth through *postnatal week 2* in mice (28, 111), but in humans alveolarization begins in the third trimester and persists for up to 3 years postnatally (FIGURE 1) (61, 130). Of note, environmental stimuli can have profound effects on the fetal lungs during development, such as oxygen availability (73, 163). For example, exposure to smoke during pregnancy has been shown to induce hypoxia in the fetus, which can have long-term consequences on lung development and function after birth, including a reduction in the number of alveoli (73, 98) and smaller lung size (81, 147).

Although the development and cellular components of the lung epithelium has been well characterized, the developing lung also requires proper formation of extensive branched vascular and peripheral nervous networks in addition to numerous mesenchymal cell types. Although the diversity and function of all mesenchymal cell types in the

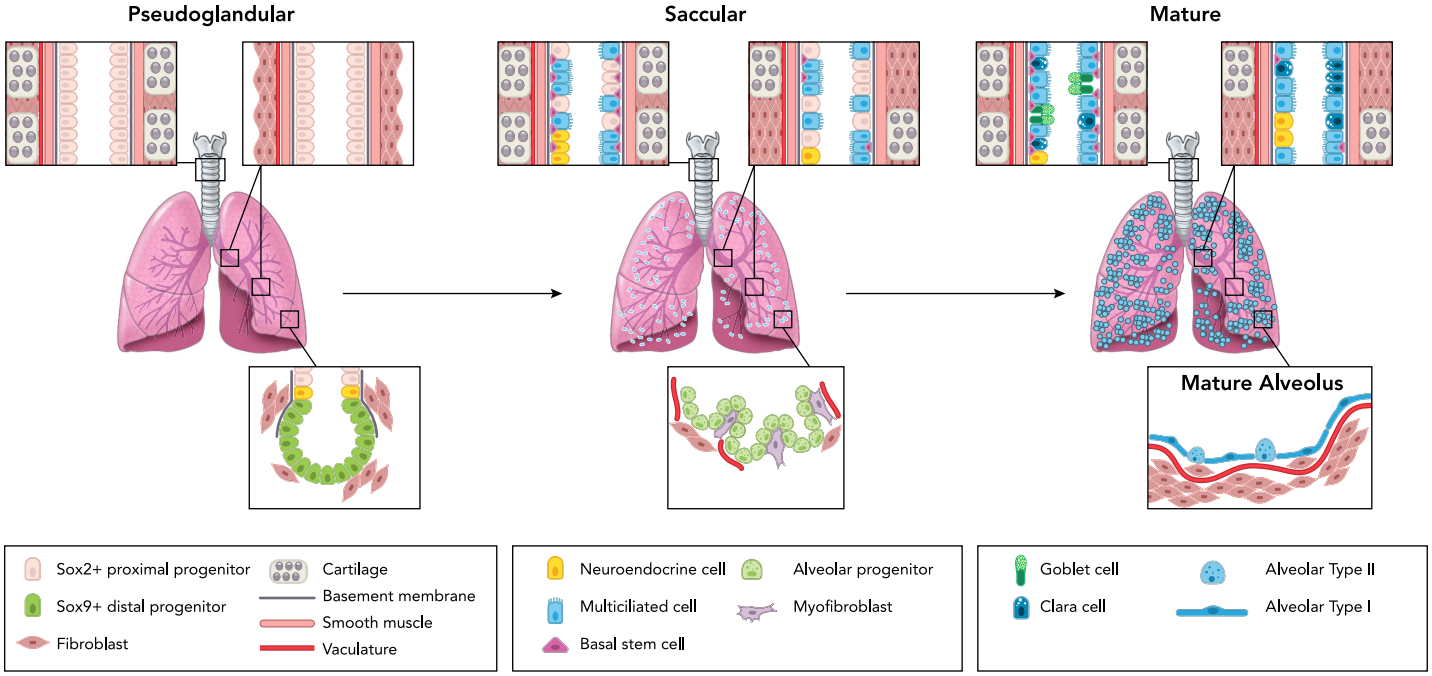


FIGURE 1. **Development and anatomy of the human lung**
Left: branching morphogenesis occurs during the pseudoglandular stage of lung development. At this time point, the proximal epithelial regions contain Sox2+ progenitor cells, whereas distal epithelial bud tips contain Sox9+ progenitors. As development progresses, Sox9+ distal bud tip progenitors give rise to alveolar-specific progenitor cells, and rudimentary saccules form (*middle*). Additionally, the proximal airway begins to form mature ciliated and secretory cells at this time. As the lung matures, saccules become mature alveolar structures comprised of AECI and AECII cells (*right*).

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lung are still poorly understood (29, 63, 102), we are beginning to understand the importance of vascular and neuronal networks during development. When normal vascular development is disrupted, for example, due to congenital diaphragmatic hernia or due to ventilator-induced lung injury in preterm infants, severe pulmonary hypertension and oxygen insufficiency can develop, significantly increasing morality rates (4, 14, 161). The vascular endothelial cells themselves have been shown to play an important role in cell-cell signaling to drive epithelial cell growth and differentiation during development and regeneration (69, 93, 158). Similarly, during branching morphogenesis, neural crest cells migrate throughout the branching lung buds where they form neurons that are closely associated with the developing bronchial tree. Disruption of these bronchial neurons leads to severe defects in branching morphogenesis (20, 24, 50).

Another critical aspect of lung development that has been difficult to study in vivo is the role of biomechanical forces and the extracellular matrix (ECM) (122, 166). Although little is known in humans, mouse studies have shown that the basement membranes surrounding branching epithelial structures have dynamic ECM remodeling that is essential for proper lung development (72, 141, 176). Similarly, three-dimensional (3D) imaging of developing mouse lungs suggests collagen and elastin networks are deposited and extensively remodeled by mesenchymal cells in close association with alveolar epithelial cells as structurally mature alveoli form (21).

Adult Lung Homeostasis and Disease

Proper lung homeostasis and response to injury is critical for adult survival. The lungs are continuously exposed to the external environment and are vulnerable to damage from environmental toxins or contact with numerous microbes, including pathogens (79). To handle these external stimuli, the adult lung must be able to clear debris, mount immune/inflammatory responses, regulate cell turnover, and facilitate proper repair after damage without disrupting oxygen absorption. To carry out these diverse functions, the lung contains specialized epithelial cell types in the proximal airway, distal alveoli, and mesenchymal compartments that are critical to lung homeostasis and reaction to injury (FIGURE 1).

In both mice and humans, the proximal airway epithelium contains numerous specialized cells that participate in normal function and cell turnover as well as repair after injury. The proximal airway is primarily comprised of specialized cells that secrete mucous to trap environmental debris [Goblet cells (68)] or that aid in innate immunity by secreting antimicrobial peptides [Club cells (15)], and cells with multiple cilia that beat in unison to remove mucous from the airway. Disruption of mucociliary function, from genetic or acquired defects, can lead to repeated chest infections and airway remodeling in humans (89, 159, 177). Mucociliary clearance and the antimicrobial properties of the lung are severely impaired in patients with cystic fibrosis (CF) (34, 78, 149). Additionally, roughly 1% of cells in the proximal airway are neuroendocrine cells, which can occur alone or in small clusters of neurons and peptide-producing cells called neuroendocrine bodies (NEBs) (16). Disregulation of NEBs has long been noted in many infant and adult pulmonary diseases (35, 66, 124), and recent evidence in mice has shown that disruption of NEB formation results in a large immune response and irreversibly simplifies alveoli (22). In addition to these cell types, the most proximal airways (the trachea only in mouse, and the trachea and primary and secondary bronchi in humans) contain basal stem cells on the basolateral side of the epithelium. These stem cells can give rise to any of the cell types of the proximal epithelium (80, 112, 140) and will be discussed further in the context of lung regeneration.

In contrast to the proximal epithelium, the adult alveoli contain specialized cell types that are optimized to facilitate gas exchange. The alveolus is a balloon-like structure lined by long, thin alveolar epithelial type I cells (AECI), which provide a thin barrier between the air and the surrounding microcapillary network to allow efficient diffusion of gas. Each alveolar sac also possesses alveolar epithelial type II cells (AECII) that secrete surfactant proteins to reduce surface tension, preventing airway collapse during exhalation. Diseases affecting the alveoli are especially lethal, since they directly interfere with gas exchange. Although specific cell types of the alveoli can be regenerated (7, 40, 167), so far there is little evidence that, once the structure of the alveolus is lost, it can be reinstated. The complex pathological processes that give rise to adult distal lung diseases remain poorly understood. In simplified terms, chronic obstructive pulmonary diseases (COPD) such as emphysema may be caused by an increased activity of enzymes that destroy the ECM of the alveoli (8). In interstitial lung disease (ILD), oxygen diffusion is disrupted by an increased deposition of ECM proteins between alveolar cells and the surrounding capillary network, increasing the distance gas needs to diffuse to enter the bloodstream (9, 60, 150).

In addition to the epithelial cells of the lung, diverse populations of mesenchymal cells surround the bronchial tree. The properties and

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functions of these populations are still an area of intensive study (63). Smooth muscle fibers surround the most proximal airways and function to regulate airway constriction in response to allergens. Thickening and hyperresponsiveness of airway smooth muscle are classic hallmarks of asthma (45). The lung mesenchyme also contains extensive vasculature, which branches in concordance with the bronchial tree and supplies and maintains the capillary network that is essential for gas exchange. Compromised endothelial cell function can lead to increased vasoconstriction and inflammation, subsequently leading to pulmonary arterial hypertension, a disease that can result in heart failure and death (144).

Adult Lung Regeneration

The adult lung is largely quiescent (134) but has a remarkable capacity to regenerate after injury (10). To date, both animal models and some in vitro models using mouse- and human-derived tissue have begun to uncover the mechanisms regulating the proliferation of lung progenitor and stem cells in the upper airways and alveoli (40, 76, 86, 91, 121, 139, 174), and have begun to uncover critical roles for mesenchymal cells in regulating repair after injury (29, 123).

The proximal airway contains basal stem cells, which are able to self renew and give rise to secretory, ciliated, and neuroendocrine cells (80, 139, 140). In addition to basal stem cells, significant cellular plasticity allows other cell types in the epithelium to act as progenitor cells and contribute to homeostasis by proliferating and repopulating lost cells after injury. For example, upon ablation of mouse basal stem cells, luminal secretory cells are able to de-differentiate into stable basal stem cells (157). Similarly, secretory club cells are able to generate new ciliated cells after injury, a process referred to as transdifferentiation (133).

Similar to the proximal airway, the distal airways and alveolar regions contain both putative, dedicated stem cells and a population of cells that are able to transdifferentiate to regenerate the epithelial layers after injury. In the mouse, bronchio-alveolar stem cells (BASC) reside at the junction between the distal airway and the alveolus and can give rise to multiple cell types upon injury (86). However, it is unclear whether human lungs contain BASCs. Additionally, in both the mouse and the human, AECII cells in the alveoli have been shown to act as an alveolar progenitor, proliferating to replenish lost AECII cells after injury, and giving rise to AECI cells in mice (7, 40). Interestingly, AECI cells have also been reported to give rise to AECII cells in the mouse after injury (85). Recent evidence suggests that there are also rare stem/progenitor cells in the distal lung that do not express mature lineage markers [lineage negative epithelial progenitors (LNEP)] and that become active in response to severe distal lung injury in mice (167). Tissue sections from patients with severe lung fibrosis were phenotypically similar to those of mice in which LNEPs were induced during injury repair, suggesting a similar population of cells and similar repair mechanisms may be at play in human disease or injury repair.

The homeostasis and regenerative capacity of diverse mesenchymal populations of the lung remains poorly characterized, although we are beginning to understand the complex cross talk between the mesenchyme, epithelium, and vasculature during lung regeneration. For example, both lipofibrobasts and myofibroblasts proliferate after injury to repopulate lost cells, and evidence suggests that lipo- and myofibroblasts also play a role in repairing the alveolar compartments after injury (29). Interestingly, a recent study utilizing genetic lineage labeling in mice has shown that lipogenic fibroblasts switch to a myogenic fibroblast phenotype during the formation of fibrosis, and a reverse myogenic to lipogenic transition is seen during the resolution of the fibrotic injury. Further experiments utilizing human lung tissues confirmed that this fate switching of fibroblast populations might be involved in the generation of human IPF (1). Similarly, the role of vascular endothelial cells in directing repair is an area of active investigation, and work has shown that cultured mouse endothelial cells supported BASC self-renewal and differentiation in vitro (93). Together, these studies highlight the importance of mesenchymal and endothelial populations on lung homeostasis and disease, and underline our current lack of understanding of their roles in development and disease mechanisms.

A Need for Human Lung Models

Most of what we know of adult human lung homeostasis is inferred from rodent studies or is gleaned from tissue samples obtained from human lungs. Studies utilizing human samples for analysis are inherently difficult to interpret, since it is difficult to control for lifestyle and genetic variability in different subjects. Similarly, despite the fact that rodents and human lungs contain most of the same cell types, the anatomy of the lung varies. For example, as elaborated above, human lungs contain basal cells throughout the trachea and bronchi, whereas mouse basal cells are restricted to the trachea; there is no evidence that human lungs possess a BASC population; the majority of cells in the human proximal airway

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are multiciliated cells, whereas club cells are more abundant in rodents (105); and, although goblet cells are prevalent in the proximal human airway, they primarily appear in the mouse after injury (120). In vitro models of lung homeostasis should be particularly powerful for increasing our understanding of how the human lung maintains homeostasis and how disregulation of specific cellular processes lead to disease, especially in hard-to-study lung regions like the human alveolus. It also may allow for tightly controlled experiments exploring the direct roles of mechanical force, immune interactions, and the ECM on progenitor differentiation and response to injury. Additionally, leveraging in vitro models could aid the discovery of novel therapeutic targets, may provide powerful, scalable screening platforms to test pharmaceuticals, and can act as an important preclinical step that bridges the gap between drug testing in rodent models and human clinical trials, which are expensive and have a high failure rate.

What Makes a Good Lung Model?

The human lung undergoes a complex developmental program, maintains homeostasis in the adult to allow efficient gas exchange, and defends and repairs itself in response to continuous outside assaults, including inhaled environmental and infectious agents. Mechanistically, influences such as biomechanical forces, availability of oxygen, morphogen signaling, and molecular-level regulation of gene and protein expression work together to regulate lung development, homeostasis, and regeneration (21, 26, 76, 87, 111, 166). Currently, in vitro models utilizing human cells from cell lines, donors, or tissue derived from human pluripotent stem cells (hPSCs) can reproduce some of these developmental events; however, these models may represent only a partial picture of processes that are more complex in the native lung. On the other hand, current in vitro systems offer tightly controlled cellular environments that can be evaluated in real time, and easily scaled and reproduced, offering significant advantages over animal models or clinical studies. As the field moves forward, layers of complexity can be added to these reductionist systems to model increasingly complex biological interactions, for example, by coculturing epithelium with vascular, neural, or immune cells, or using engineering approaches to develop more complex matrix environments (59, 179), which may begin to more closely mirror the environment of the native lung.

In Vitro Models Utilizing Cells Derived From the Human Lung

Immortalized Human Cell Lines

Human cell lines have been a simple and powerful tool for studying lung cell biology for decades, and remain popular especially in the CF field (51, 65), where access to primary tissue from patients can be limited. Cell lines have been cultured at an air liquid interface (ALI) to induce differentiation of cells with mucociliary function, and have been used to show that CFTR mutations do not directly affect neutrophil migration (128) and to study the effect of CFTR mutations on cell permeability and cell-cell junction integrity (107, 171). In addition, cell lines with mRNA expression patterns similar to AECII cells have been used to probe the mechanisms governing fluid transport across the alveolar epithelium (137). Transwell coculture systems, utilizing a lower chamber in which human vascular cells are grown on one side and cell lines derived from human airway or alveolar epithelium are grown on a permeable membrane, have been used to study epithelial resistance and barrier function (75). Furthermore, a cell line composed primarily of basal-like cells has been used to form self-assembling organoids in a 3D ECM environment (49). Coculture of these cells with human vascular endothelial cells (HUVECs) in a 3D environment promoted morphological rearrangements similar to branching morphogenesis, leading to domains with both proximal and distal cell types. This work suggested that proximal airway cells may have the capacity to generate distal cell types given the correct environment in culture.

Since human cell lines are immortalized or transformed, it is unclear how well these cells maintain the physiology of normal airway cells, which is a major limitation to the usefulness of these cells to generating complex in vitro models of the human lung. In this light, more recent work has focused on the use of non-transformed primary lung tissue from human donors to model lung physiology, utilizing a variety of culture approaches. Adult lung epithelium has been grown in monolayers, on an ALI, in self-organizing 3D environments (organoids), and in a variety of bioengineered or decellularized biological scaffolds to provide models of the upper and lower airways.

Monolayer and ALI Culture Systems

Some of the earliest work growing primary human lung cells in culture employed a two-dimensional (2D) approach, where primary epithelial cells taken from human nasal or bronchial brushings were cultured on a plastic plate submerged in media (143). 2D monolayer cultures are simple to generate and are used clinically to diagnose ciliary

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dysfunction (127, 160). The development of the ALI method in the late 1980s was a major advancement in the field (172). In the ALI system, cells are grown on a porous filter that physically separates the lung epithelial tissue from the underlying media, allowing the surface of the cells to interface with the surrounding air, while the basal surface of the cells have access to nutrients and other media additives via diffusion through the porous membrane (FIGURE 2). When human nasal or bronchial epithelial cells are grown on the ALI, ciliated and mucous-producing cells become polarized and take on the proper apical-basal morphology, including functional apical cilia and/or mucous secretion from goblet cells (6, 172, 180). Additional advances such as the addition of Rho-associated protein kinase (ROCK) inhibitors, "dual smad" inhibition, or the coculture with fibroblast feeder cells have improved the ability to culture basal stem cells long-term and for many passages (12, 52, 95, 112, 154). ALI cultures of primary human epithelial tissue have been particularly instrumental in understanding cellular mechanisms of CF, where these systems have been used to study the cellular effects of CFTR mutations or defects in epithelial barrier function (94, 107, 116, 131) and have been instrumental in drug discovery to treat certain subsets of CF (38, 164). ALI cultures have also been employed to evaluate the effect of external toxins such as cigarette smoke, environmental particles, or infectious agents such as HIV and influenza on bronchial epithelial cells (23, 33, 145), and to assess the role of soluble signals from healthy or diseased feeder fibroblasts on the gene expression and behavior of epithelial cells (119).

Self-Assembling 3D Cultures

More recently, primary adult human lung tissue has been cultured in 3D environments, which leads to self-assembly of the tissue into a variety of forms that partially reflect the structure of the lung (37, 140, 146, 156).

The generation of human "bronchospheres," 3D spheres derived from primary human bronchiolar epithelial cells grown in a 3D matrix, is a fairly recent innovation in the field (140). This method involves isolating basal stem cells from mouse or human epithelial tissue. When these cells are embedded in a 3D ECM gel, spherical clonal colonies form after a short time in culture. These initial experiments provided a platform to conduct functional experiments in human tissue to show that human basal stem cells are able to self-renew and give rise to proximal secretory and ciliated cells (140). More recently, human bronchospheres were used as a screening platform to probe the epithelial response to specific stimuli, identifying a number of proteins that bias basal stem cell differentiation toward a secretory cell fate (37, 77). Although several reports have shown the generation of epithelium-only 3D bronchospheres containing cell types of the proximal or distal airway, bronchiolar

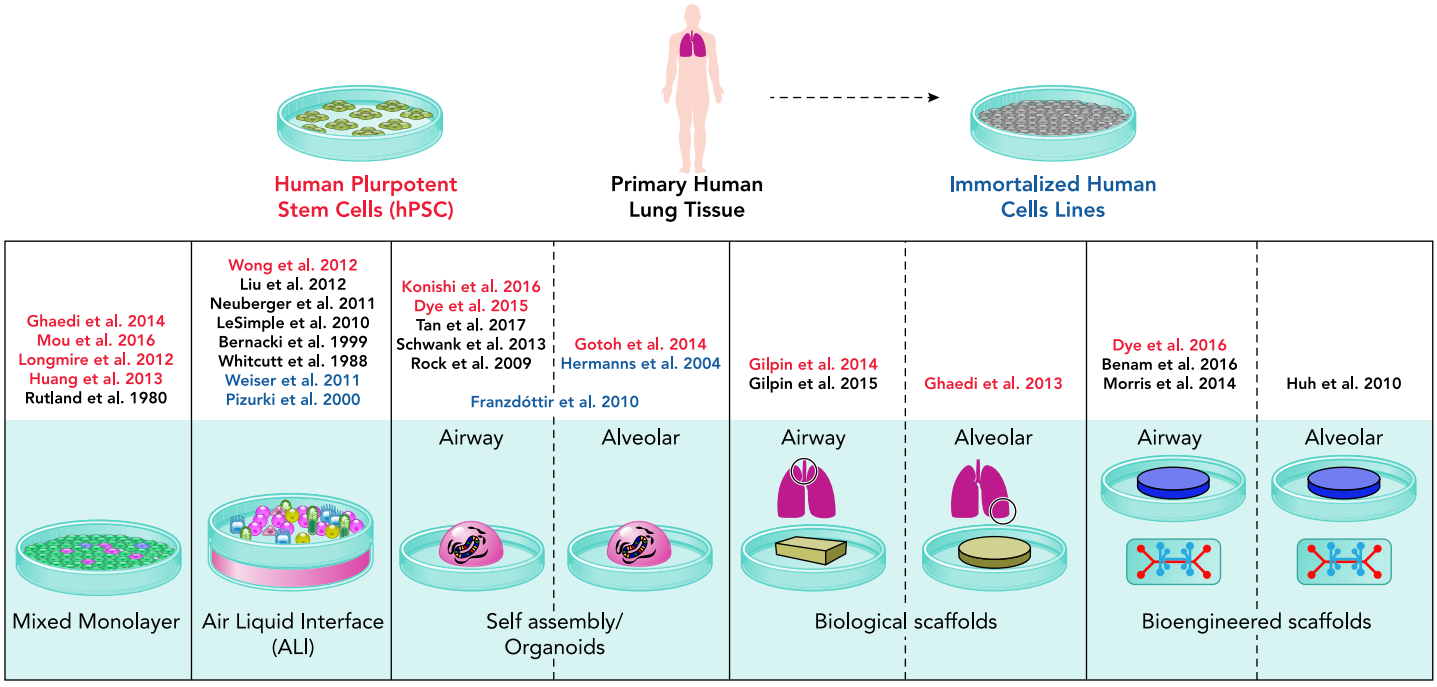


FIGURE 2. **In vitro models of the human lung**
A variety of in vitro human model systems have been developed using primary lung tissue (examples denoted in black text), lung cells from immortalized human cell lines (examples denoted in blue text), and lung-specific cells derived from hPSCs (examples denoted in pink text). These systems have been developed in a variety of contexts, including monolayer, ALI, or a variety of 3D and engineered environments to recapitulate elements of the native human lung.

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epithelium has also been cocultured with lung fibroblasts and endothelial cells (156). Interestingly, these cultures rapidly formed cell condensations that had properties of both proximal and distal airway compartments, suggesting that proximal cell types maintain a level of plasticity that can be reactivated in organoid culture, perhaps due to specific cues from the microenvironment provided by cocultured cells (156). Interestingly, it has been shown that mouse AECII cells are able to give rise to "alveolospheres" when placed in culture, suggesting that AECII cells are stem cells in the adult lung (barkauskas, Hogan JCI 2013). Of note, growth of primary human alveolospheres has not yet been reported. In vitro coculture of mouse epithelial progenitor cells along with mouse endothelial or fibroblasts has been successfully carried out and may be leveraged in the future to identify mechanisms that support certain cell populations or methods for selectively isolating specific cells from whole lung tissue (74, 104).

3D culture systems utilizing primary human cells are especially amenable to studying the proliferative and stem-like properties of cell populations in colony formation assays and for studying patient-specific disease states. For example, organoids generated from colon biopsies of CF patients have been used in a forskolin-induced swelling assay to study functionality of the CFTR in individual patients (146).

Biological Scaffolds

Although isolated human lung cells are able to survive in 2D on plastic and in 3D Matrigel or collagen environments, the ECM in the native lung is intricate and complex, and shapes the cell microenvironment by providing structural cues, regulating access to soluble growth factors and controlling cell adhesion and migration (103, 122, 168, 173). The ECM is extensively remodeled in disease states such as pulmonary fibrosis and emphysema (17). The ability to model cell-ECM interactions to better understand disease and the promise of manufacturing a new lung for patients with end-stage lung disease have led researchers to develop methods for decellularizing rodent and human lungs or lung sections and reseeding human lung cells onto the matrix. Multiple techniques have been developed to decellularize lungs (19, 57, 67, 118, 126, 169), although the ability to decellularize human lungs currently far exceeds the success of recellularizing them to generate mature, functional cell types seen in the native lung.

Early successes at recellularizing scaffolds relied on seeding them with primary mesenchymal stem cells (18), which are able to engraft and proliferate in the scaffold. More recently, basal stem cells isolated from donated human lungs and expanded in culture were seeded onto decellularized rat lungs where they maintained a basal cell identity (56). Despite advances in the creation of decellularized scaffolds, significant advances improving seeding cells on the scaffolds and improving cell survival and differentiation/maturation will be critical for creating functional recellularized lungs for organ transplant or for generating functional models of the lung to study basic lung biology. One hypothesis for the difficulty of reseeding scaffolds is a lack of supporting cells and structures, such as the vascular network. Studies have reported the successful repopulation of endothelial cells in decellularized rat (153) and human lung scaffolds (138),which led, in both cases, to a partial reestablishment of barrier function. Moving forward, the successful inclusion of multiple supporting cell types including endothelial, mesenchymal, and neuronal cells, reseeded in the proper niche, may prove critical for improving successful recellularization of whole organs.

Bioengineered Scaffolds and Niches

Although decellularized scaffolds are a powerful tool, the stiffness, protein/carbohydrate composition, and residual growth factors within scaffolds can vary among species. Furthermore, the composition of scaffolds is dependent on the decellularization method, which can lead to significant differences in the properties of the native matrices and in cell-seeding efficiency (5). In an effort to create a welcoming environment for human lung cells to thrive, researchers have turned to bioengineering approaches to create a variety of structural scaffolds and microenvironments to reproducibly mimic the lung. Several studies have attempted to seed primary lung cells from mice or rats onto scaffolds comprised of naturally occurring ECM, such as Matrigel (3, 108) and collagen (117), or on synthetic materials including polyglycolic acid (PGA), polylactic-coglycolic acid (PLGA), and pluronic F-127 (31, 129, 162). For example, novel manufacturing techniques have been reported to create a thin wafer composed of nanofibers of varying diameter, which structurally resemble the basement membrane of the bronchial epithelium. The authors were able to successfully coculture human bronchial cells and fibroblasts at the ALI on these wafers, showing that the fibroblast cells invaded the structure, whereas bronchial cells grew and matured at the ALI (110). Additionally, lung mesenchymal cell types taken from fetal lung tissue or derived from hPSCs have been seeded onto alginate beads coated in collagen and cultured in a bioreactor, which resulted in the formation of mesenchymal organoids that have a structural phenotype similar to the alveolar regions of the native lung (175). Although seeding cells on scaffolds is an

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excellent tool for probing questions related to the effect of the ECM on cell differentiation and repair, these models may not recapitulate structural or mechanical forces that are constantly at play in the breathing lung.

To model mechanical forces of the lung such as stretch, bioengineers have developed innovative microsystems, called "lungs-on-a-chip," in which the location and mechanical force placed on each cell are tightly defined to model both the small airway and the stretch experienced in the alveolar compartment (11, 84). In the alveolar lung-on-a-chip, structural, functional, and mechanical elements of the interface between the alveoli and the capillary network were modeled by microfabricating a microfluidic device that contains two channels separated by a thin flexible membrane coated with ECM proteins. Vascular endothelial cells were cultured in one chamber, and alveolar epithelial cells in the other, which could also be exposed to air. Controlled regulation of stretching of the membrane and fluid flow through the vascular chamber allowed study of the passage of materials and nutrients from the alveolar region to the vascular compartment (84). Follow-up work with this device has shown that mechanical forces generated during breathing can play a role in the development of vascular leakage associated with pulmonary edema in response to interleukin-2 (IL-2) treatment (83). More recently, a "small-airway-on-a-chip" has been developed, which houses a bronchial epithelium with mucociliary differentiation in one chamber exposed to air, and with the vascular endothelial layer in the opposite, fluid-filled chamber. Proof-of-concept experiments with this device showed that treating cells with IL-13 resulted in goblet cell hyperplasia and decreased cilliary function. Additionally, when cells from patients with COPD were used, the system was able to recapitulate some aspects of the disease state (11). To date, these model systems are the only systems to incorporate controlled mechanical stretching and liquid flow control to recapitulate physiological function at the organ level.

Benefits and Drawbacks of Lung Models Utilizing Primary Human Cells

Systems to culture primary human lung cells have promoted investigation into mechanisms regulating homeostasis, repair, and disease in the upper airway and, to some extent, the alveolus. In vitro models employing primary tissue provide certainty that the cell types obtained are biologically relevant and are patient specific, which overcomes one of the major limitations of using immortalized cell lines. Recent advances in coculture conditions with mesenchymal cells and with various types of ECM components have made these systems more attractive as models for studying complex interactions between cell types in chronic lung disease, such as COPD, ILD, or cystic fibrosis. It would also be possible in this context to generate models from individuals with chronic diseases such as CF that include elements of the diseased microenvironment of the lung, including the general microbiota and/or multiple strains of antibiotic-resistant bacteria, which may have drastic implications for how well that patient responds to certain drug therapies.

Despite advantages, these systems also have caveats and drawbacks. For example, in vitro grown cells are supported by an artificial niche, which may not closely model the in vivo environment. Availability of primary human tissue is limited, and, in most cases, each laboratory will have unique cell lines. Thus the use of non-standardized cell lines makes it difficult or impossible to control for donor heterogeneity as a biological variable and also renders these systems ill suited for high throughput studies like drug screens. Recent advances in differentiating human pluripotent stem cells to specific lung lineages, discussed below, have alleviated some of the concerns over donor heterogeneity and tissue availability; however, the use of primary human lung cells will no doubt continue to play an important role in evaluating disease and patient-specific physiology.

In the future, building complexity within in vitro lung models in an attempt to provide a more comprehensive picture of lung function will increase the utility of these systems. For example, developing systems that include epithelial, neural, mesenchymal, endothelial, and human immune cells will provide increased biological complexity in a defined and experimentally tractable environment. This may be particularly useful for diseases like IPF and COPD, where better understanding of epithelial-mesenchymal interactions may help improve our understanding of disease.

Directed Differentiation of hPSCS to Generate In Vitro Human Lung Models

Another area of scientific inquiry where there has been a flurry of progress developing in vitro models of the human lung has been in the field of human pluripotent stem cells (hPSCs), which include both embryonic and induced pluripotent stem cells (43, 54, 62, 71, 82, 90, 96, 113, 178). Most approaches to differentiate hPSCs into lung lineages follow an experimental paradigm called directed differentiation (4, 44). Directed differentiation aims to mimic in vitro the signaling cues that control cell fate decisions in the embryo in a step-by-step fashion. Although slightly different

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methods exist, most of the studies differentiating hPSCs into lung lineages share a general experimental framework. That is, hPSCs are coaxed to differentiate in a stepwise process through several cellular states that mimic gastrulation, patterning of the lung field, and specification of a lung lineage. In the case of the lung epithelium, this includes induction of the definitive endoderm (gastrulation), followed by anterior foregut endoderm (patterning), and finally induction of a lung identity (specification) (reviewed in Refs. 44, 55, 151). To date, researchers have created hPSC-derived lung cell models in many of the same contexts as primary human epithelial tissues discussed above, including 2D culture models, ALI cultures, 3D environments, and seeding cells onto biological or synthetic matrices.

Monolayer Cultures

The ability to differentiate lung lineages from hPSCs was built on seminal work demonstrating how these cells could be instructed to generate endoderm (mimicking gastrulation), followed by foregut specification (mimicking embryonic patterning) (36, 64). Thus the first reports of lung lineage differentiation from hPSCs took place in a 2D environment and gave rise to progenitor cells that were similar to those induced early during lung development in the embryo (96, 113). To mature these hPSC-derived progenitors, culture conditions were identified that allowed many cells within the culture to stochastically differentiate into lung-epithelium-specific cell types, including those from both the airway and the alveoli (82). However, 2D cultures generally show a lack of organization, with different cell types randomly distributed throughout the culture. To achieve an enriched airway-like cell population of lung progenitors, hPSC-derived lung progenitors were cultured at the ALI for many weeks. This promoted differentiation of multiciliated, goblet and basal cells, including some functional beating ciliated cells (178). Although the majority of studies generating lung lineages from hPSCs have shown success generating proximal airway cell types, differentiation of AECI-like and AECII-like cells has also been reported (54, 62, 82). In one report, directed differentiation was used to generate AECI- and AECII-like cells, which were then seeded on a decellularized scaffold to generate more mature alveolar cell-like phenotypes (53). Additional studies showed that alveolar progenitor cells could also be matured in vitro by culturing 2D iPSC-derived in a bioreactor that alternately exposed cells to the air and culture media (54).

The use of patient-specific iPSCs has been a boon to research on genetic diseases affecting the lung, particularly CF. Mature lung cells from CF iPSCs were recently generated (113). Multiple groups have used CF patient-specific iPSC-derived lung cells to show that genome engineering approaches can be used for functional gene correction to the CFTR in a patient-specific manner (32, 48), potentially paving the way for the use of these systems in identifying novel therapeutic approaches, such as gene corrected patient-specific tissue replacement.

Self-Assembly

Although 2D cultures are well suited for studying epithelial barrier function and understanding biochemical components of disease, 3D culture systems have the added advantage of beginning to recapitulate some of the complex structural environment of the native lung. By modifying the differentiation conditions, endoderm cultures can be directed to differentiate into lineage-restricted 3D structures, including foregut "spheroids" (43, 70, 100, 101, 152), which can be grown in a 3D ECM droplet such as Matrigel. Spheroids can be expanded into "human lung organoids" (HLOs) by adding growth factors such as FGF10, which are important for embryonic lung development (43, 111, 148). HLOs possessed proximal airway-like structures, including basal cells and poorly differentiated ciliated cells, but lacked secretory-like cells. HLOs included regions containing cells expressing markers of AECI and AECII cells, but generally lacked alveolar structure. Additionally, proximal epithelial structures within HLOs were surrounded by lung mesenchymal cell types, including smooth muscle cells (43). Follow-up work in this system showed that HLOs were able to give rise to more mature, adult-like structures when transplanted into mice (42). Generation of hPSC-derived bronchospheres has also been reported (90), where hPSC-derived lung progenitor cells were purified from 2D cultures and placed in a 3D ECM, which led to the formation of epithelium-only cysts. Treatment of the cysts with DAPT, a notch inhibitor, induced the formation of neuroendocrine cells, mucus-producing goblet cells, and functional multiciliated cells (90).

Although the formation of proximal airway-like structures and cell types has been reported in a number of contexts, there are fewer reports of the differentiation of alveolar cell types (71, 82), and only one report has demonstrated clear evidence of alveolar cell-type differentiation in a 3D context (62). hPSC-derived lung progenitors were purified using a cell-sorting strategy and cocultured in a 3D environment with primary fetal human lung fibroblasts, which resulted in the generation of AECI- and AECII-like alveolar epithelial cell types (62).

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Biological and Synthetic Scaffolds

Many researchers have turned to biological and synthetic scaffolds to attempt to mature hPSC-derived lung progenitors into functional adult cell types. For example, hPSC-derived AECII-like cells can engraft and proliferate on decellularized rat and human lung scaffolds, and these cells expressed markers of mature epithelial cell types after seeding (53). Similarly, hPSC-derived lung progenitors have been shown to seed onto decellularized lung slices or whole organs (58), which is a first step in the effort to generate functional whole organs from patient-specific iPSCs for transplantation. A bioengineered synthetic scaffold was used to enhance engraftment and growth of hPSC-derived lung spheroids in mice, leading to tube-like airway structures with basal cells, functional ciliated cells, secretory cells, and many mature mesenchymal cell types surrounding the epithelial tubes (42).

Benefits and Drawbacks of Stem Cell-Derived In Vitro Lung Model Systems

The use of hPSCs to model human lung development and disease represents an enormous potential for generating patient-specific models and for overcoming barriers of tissue availability, donor heterogeneity, and tissue quality. An ideal stem cell-derived system would be robust and reproducible with the potential for scale-up for cell therapy or large drug screens. Despite these advantages, the field has a number of areas that need to be improved to maximize the usefulness of these model systems. Although many reports use similar strategies to generate lung progenitor cells, no consensus has been reached for a standardized method, and, therefore, small differences persist. For example, subtle differences in the growth factors used can lead to drastic changes in experimental outcomes. Additionally, hPSC-derived tissues tend to remain immature, similar to fetal tissue, and do not generally mature to an adult-like state in vitro (4, 43, 46, 101, 155, 170). Thus more work needs to be done to understand how to mature hPSC-derived lung tissue and to demonstrate definitively that stem cell-derived lung cells have the same functionality as well as similar chromatin states and gene expression profiles as their in vivo counterparts before meaningful conclusions can be drawn from in vitro studies of adult cell behaviors.

Future Directions

Significant progress has been made in the past decade to develop new in vitro models of the human lung. However, many major biological questions and technical challenges remain. For example, hPSC studies have led to methods that promote differentiation of multiple lung epithelial cell types at the same time. This is, in part, because current methods for deriving lung tissue from hPSCs all use slightly different combinations of growth factors, and all generate heterogeneous populations of cells. Future studies should ideally use a more systematic approach to untangle the mechanisms that control differentiation of specific cell lineage decisions in vitro. This will lay fundamental groundwork that can be used to selectively differentiate more homogenous populations of cells. This will both improve the fundamental understanding of lung lineage cell fate choice and be advantageous for more controlled tissue engineering and translational applications of these cells.

Another remaining question in the field is whether hPSC-derived lung cells can fully recapitulate the function and phenotype of mature in vivo lung cells in vitro, while avoiding transplantation into an animal to reach maturity. Studies in the lung, and in other hPSC-derived tissues such as intestine, stomach, kidney, and brain, have shown that in vitro organoid tissues are similar to their in vivo fetal counterparts (4, 25, 44, 47, 100, 155, 170). In vivo transplantation of hPSC-derived cells or tissues into immuocompromised mice promotes molecular, morphological, and functional maturation (42, 46, 92, 179). However, the signals that prompt maturation following transplantation are completely unknown. Although this is a relatively daunting problem to approach experimentally, one way forward will be to ask which factors from the host are responsible for maturation, such as blood flow, immune factors, neural innervation, or the mechanical strain of the local environment. It is likely that a combination of experimental approaches, including surgical, engineering, and genetic approaches, will allow the modulation of some of the variables in the host that may play a role in tissue maturation.

Although the majority of the systems discussed in this review have attempted to model either proximal (airway) or distal (alveolar) cell types, few systems are able to integrate both compartments of the lung into one model and introduce additional complexity, such as the mesenchymal, vascular, immune, and neural tissue. It is abundantly clear from the literature that interaction between these distinct anatomical compartments and unique cell types all play critical roles in guiding lung development, homeostasis, and pathogenesis. Taking calculated steps to merge two or more cell types at a time in 3D cultures will be instrumental in developing reproducible and modular systems that can more faithfully recapitulate development or disease. However, merging these complex

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systems together will likely come with additional challenges. For example, the physical environment in which different cell/tissue types are grown in vitro can dramatically affect cellular function. Epithelial tissues, for example, grow best in laminin-rich matrices (such as Matrigel), whereas endothelial and mesenchymal cells tend to grow more robustly in collagen. These differences will present additional technical hurdles that will need to be overcome.

In addition to the potential for in vitro model systems to inform our understanding of the effect of growth factors, oxygen availability, or other environmental factors to control cell fate and tissue morphology, one area where in vitro systems may prove to be extremely powerful is in the investigation of how the physical environment influences lung biology and disease. This includes investigating the role of the ECM, including the tensile environment that is generated by mechanical forces in the lung. Notable recent work has shown that the mechanical environment alone is capable of inducing branching of the lung epithelium in vitro (165) and that the restrictive forces generated by smooth muscle development play a critical role in branching morphogenesis (88). Additionally, exciting recent work engineering biomimetic, chemically defined, and fully synthetic hydrogels to grow complex tissues such as organoids has illuminated a path to begin to address questions about how the physical environment may regulate cell fate and tissue homeostasis (59, 115). This type of defined system will allow precise control over the extracellular environment to ascertain which elements of the ECM or other physical parameters (e.g., stiffness, stretch) are essential for development and morphogenesis or which may influence disease states. Another major hurdle for the field is to generate methods to maintain physiological shear stress in culture systems and to efficiently model the mechanical environment of the developing lung as well as the homeostatic and pathological mature organ states. Developing a deep and clear understanding of how the physical and chemical environment control progenitor cell fate decisions, maturation, function, response to injury, and tissue homeostasis both in vitro and in vivo is also a major challenge. In this light, it should be pointed out that, despite the power of in vitro human models, in vivo animal models will be an integral part of continuing to build an understanding of lung biology. In the future, cooperation among bioengineers, mathematical modelers, and biologists will be essential to develop a comprehensive understanding of how the complex structural environment of the human lung promotes health or leads to disease. ■

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References

1. El Agha E, Moiseenko A, Kheirollahi V, De Langhe S, Crnkovic S, Kwapiszewska G, Kosanovic D, Schwind F, Schermuly RT, Henneke I, MacKenzie B, Quantius J, Herold S, Ntokou A, Ahlbrecht K, Morty RE, Günther A, Seeger W, Bellusci S. Two-way conversion between lipogenic and myogenic fibroblastic phenotypes marks the progression and resolution of lung fibrosis. *Cell Stem Cell* 20: 261–273.e3, 2017. doi:10.1016/j.stem.2016.10.004.

2. Alvarado A, Arce I. Metabolic functions of the lung, disorders and associated pathologies. *J Clin Med Res* 8: 689–700, 2016. doi:10.14740/jocmr2668w.

3. Andrade CF, Wong AP, Waddell TK, Keshavjee S, Liu M. Cell-based tissue engineering for lung regeneration. *Am J Physiol Lung Cell Mol Physiol* 292: L510–L518, 2007. doi:10.1152/ajplung.00175.2006.

4. Aurora M, Spence JR. hPSC-derived lung and intestinal organoids as models of human fetal tissue. *Dev Biol* 420: 230–238, 2016. doi:10.1016/j.ydbio.2016.06.006.

5. Balestrini JL, Gard AL, Gerhold KA, Wilcox EC, Liu A, Schwan J, Le AV, Baevova P, Dimitrievska S, Zhao L, Sundaram S, Sun H, Rittié L, Dyal R, Broekelmann TJ, Mecham RP, Schwartz MA, Niklason LE, White ES. Comparative biology of decellularized lung matrix: implications of species mismatch in regenerative medicine. *Biomaterials* 102: 220–230, 2016. doi:10.1016/j.biomaterials.2016.06.025.

6. Bals R, Beisswenger C, Blouquit S, Chinet T. Isolation and air-liquid interface culture of human large airway and bronchiolar epithelial cells. *J Cyst Fibros* 3, *Suppl* 2: 49–51, 2004. doi:10.1016/j.jcf.2004.05.010.

7. Barkauskas CE, Cronce MJ, Rackley CR, Bowie EJ, Keene DR, Stripp BR, Randell SH, Noble PW, Hogan BLM. Type 2 alveolar cells are stem cells in adult lung. *J Clin Invest* 123: 3025–3036, 2013. doi:10.1172/JCI68782.

8. Barnes PJ, Shapiro SD, Pauwels RA. Chronic obstructive pulmonary disease: molecular and cellular mechanisms. *Eur Respir J* 22: 672–688, 2003. doi:10.1183/09031936.03.00040703.

9. Bateman ED, Turner-Warwick M, Adelmann-Grill BC. Immunohistochemical study of collagen types in human foetal lung and fibrotic lung disease. *Thorax* 36: 645–653, 1981. doi:10.1136/thx.36.9.645.

10. Beers MF, Morrisey EE. The three R's of lung health and disease: repair, remodeling, and regeneration. *J Clin Invest* 121: 2065–2073, 2011. doi:10.1172/JCI45961.

11. Benam KH, Villenave R, Lucchesi C, Varone A, Hubeau C, Lee H-H, Alves SE, Salmon M, Ferrante TC, Weaver JC, Bahinski A, Hamilton GA, Ingber DE. Small airway-on-a-chip enables analysis of human lung inflammation and drug responses in vitro. *Nat Methods* 13: 151–157, 2016. doi:10.1038/nmeth.3697.

12. Bernacki SH, Nelson AL, Abdullah L, Sheehan JK, Harris A, Davis CW, Randell SH. Mucin gene expression during differentiation of human airway epithelia in vitro. Muc4 and muc5b are strongly induced. *Am J Respir Cell Mol Biol* 20: 595–604, 1999. doi:10.1165/ajrcmb.20.4.3442.

13. Bernhard W. Lung surfactant: function and composition in the context of development and respiratory physiology. *Ann Anat* 208: 146–150, 2016. doi:10.1016/j.aanat.2016.08.003.

14. Bhatt AJ, Pryhuber GS, Huyck H, Watkins RH, Metlay LA, Maniscalco WM. Disrupted pulmonary vasculature and decreased vascular endothelial growth factor, Flt-1, and TIE-2 in human infants dying with bronchopulmonary dysplasia. *Am J Respir Crit Care Med* 164: 1971–1980, 2001. doi:10.1164/ajrccm.164.10.2101140.

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15. Boers JE, Ambergen AW, Thunnissen FB. Number and proliferation of clara cells in normal human airway epithelium. *Am J Respir Crit Care Med* 159: 1585–1591, 1999. doi:10.1164/ajrccm.159.5.9806044.

16. Boers JE, den Brok JL, Koudstaal J, Arends JW, Thunnissen FB. Number and proliferation of neuroendocrine cells in normal human airway epithelium. *Am J Respir Crit Care Med* 154: 758–763, 1996. doi:10.1164/ajrccm.154.3.8810616.

17. Bonnans C, Chou J, Werb Z. Remodelling the extracellular matrix in development and disease. *Nat Rev Mol Cell Biol* 15: 786–801, 2014. doi:10.1038/nrm3904.

18. Bonvillain RW, Danchuk S, Sullivan DE, Betancourt AM, Semon JA, Eagle ME, Mayeux JP, Gregory AN, Wang G, Townley IK, Borg ZD, Weiss DJ, Bunnell BA. A nonhuman primate model of lung regeneration: detergent-mediated decellularization and initial in vitro recellularization with mesenchymal stem cells. *Tissue Eng Part A* 18: 2437–2452, 2012. doi:10.1089/ten.tea.2011.0594.

19. Booth AJ, Hadley R, Cornett AM, Dreffs AA, Matthes SA, Tsui JL, Weiss K, Horowitz JC, Fiore VF, Barker TH, Moore BB, Martinez FJ, Niklason LE, White ES. Acellular normal and fibrotic human lung matrices as a culture system for in vitro investigation. *Am J Respir Crit Care Med* 186: 866–876, 2012. doi:10.1164/rccm.201204-0754OC.

20. Bower DV, Lee H-K, Lansford R, Zinn K, Warburton D, Fraser SE, Jesudason EC. Airway branching has conserved needs for local parasympathetic innervation but not neurotransmission. *BMC Biol* 12: 92, 2014. doi:10.1186/s12915-014-0092-2.

21. Branchfield K, Li R, Lungova V, Verheyden JM, McCulley D, Sun X. A three-dimensional study of alveologenesis in mouse lung. *Dev Biol* 409: 429–441, 2016. doi:10.1016/j.ydbio.2015.11.017.

22. Branchfield K, Nantie L, Verheyden JM, Sui P, Wienhold MD, Sun X. Pulmonary neuroendocrine cells function as airway sensors to control lung immune response. *Science* 351: 707–710, 2016. doi:10.1126/science.aad7969.

23. Brune KA, Ferreira F, Mandke P, Chau E, Aggarwal NR, D'Alessio FR, Lambert AA, Kirk G, Blankson J, Drummond MB, Tsibris AM, Sidhaye VK. HIV impairs lung epithelial integrity and enters the epithelium to promote chronic lung inflammation. *PLoS One* 11: e0149679, 2016. doi:10.1371/journal.pone.0149679.

24. Burns AJ, Delalande J-MM, Le Douarin NM. In ovo transplantation of enteric nervous system precursors from vagal to sacral neural crest results in extensive hindgut colonisation. *Development* 129: 2785–2796, 2002.

25. Camp JG, Badsha F, Florio M, Kanton S, Gerber T, Wilsch-Bräuninger M, Lewitus E, Sykes A, Hevers W, Lancaster M, Knoblich JA, Lachmann R, Pääbo S, Huttner WB, Treutlein B. Human cerebral organoids recapitulate gene expression programs of fetal neocortex development. *Proc Natl Acad Sci USA* 112: 15672–15677, 2015. doi:10.1073/pnas.1520760112.

26. Cardoso WV. Molecular regulation of lung development. *Annu Rev Physiol* 63: 471–494, 2001. doi:10.1146/annurev.physiol.63.1.471.

27. Chang DR, Martinez Alanis D, Miller RK, Ji H, Akiyama H, McCrea PD, Chen J. Lung epithelial branching program antagonizes alveolar differentiation. *Proc Natl Acad Sci USA* 110: 18042–18051, 2013. doi:10.1073/pnas.1311760110.

28. Chao C-M, Moiseenko A, Zimmer K-P, Bellusci S. Alveologenesis: key cellular players and fibroblast growth factor 10 signaling. *Mol Cell Pediatr* 3: 17, 2016. doi:10.1186/s40348-016-0045-7.

29. Chen L, Acciani T, Le Cras T, Lutzko C, Perl A-KT. Dynamic regulation of platelet-derived growth factor receptor α expression in alveolar fibroblasts during realveolarization. *Am J Respir Cell Mol Biol* 47: 517–527, 2012. doi:10.1165/rcmb.2012-0030OC.

31. Cortiella J, Nichols JE, Kojima K, Bonassar LJ, Dargon P, Roy AK, Vacant MP, Niles JA, Vacanti CA. Tissue-engineered lung: an in vivo and in vitro comparison of polyglycolic acid and pluronic F-127 hydrogel/somatic lung progenitor cell constructs to support tissue growth. *Tissue Eng* 12: 1213–1225, 2006. doi:10.1089/ten.2006.12.1213.

32. Crane AM, Kramer P, Bui JH, Chung WJ, Li XS, Gonzalez-Garay ML, Hawkins F, Liao W, Mora D, Choi S, Wang J, Sun HC, Paschon DE, Guschin DY, Gregory PD, Kotton DN, Holmes MC, Sorscher EJ, Davis BR. Targeted correction and restored function of the CFTR gene in cystic fibrosis induced pluripotent stem cells. *Stem Cell Reports* 4: 569–577, 2015. doi:10.1016/j.stemcr.2015.02.005.

33. Creager HM, Zeng H, Pulit-Penaloza JA, Maines TR, Tumpey TM, Belser JA. In vitro exposure system for study of aerosolized influenza virus. *Virology* 500: 62–70, 2017. doi:10.1016/j.virol.2016.10.007.

34. Cutting GR. Cystic fibrosis genetics: from molecular understanding to clinical application. *Nat Rev Genet* 16: 45–56, 2015. doi:10.1038/nrg3849.

35. Cutz E, Perrin DG, Pan J, Haas EA, Krous HF. Pulmonary neuroendocrine cells and neuroepithelial bodies in sudden infant death syndrome: potential markers of airway chemoreceptor dysfunction. *Pediatr Dev Pathol* 10: 106–116, 2007. doi:10.2350/06-06-0113.1.

36. D'Amour KA, Agulnick AD, Eliazer S, Kelly OG, Kroon E, Baetge EE. Efficient differentiation of human embryonic stem cells to definitive endoderm. *Nat Biotechnol* 23: 1534–1541, 2005. doi:10.1038/nbt1163.

37. Danahay H, Pessotti AD, Coote J, Montgomery BE, Xia D, Wilson A, Yang H, Wang Z, Bevan L, Thomas C, Petit S, London A, LeMotte P, Doelemeyer A, Vélez-Reyes GL, Bernasconi P, Fryer CJ, Edwards M, Capodieci P, Chen A, Hild M, Jaffe AB. Notch2 is required for inflammatory cytokine-driven goblet cell metaplasia in the lung. *Cell Reports* 10: 239–252, 2015. doi:10.1016/j.celrep.2014.12.017.

38. Davis PB, Yasothan U, Kirkpatrick P. Ivacaftor. *Nat Rev Drug Discov* 11: 349–350, 2012. doi:10.1038/nrd3723.

39. de Bakker BS, de Jong KH, Hagoort J, de Bree K, Besselink CT, de Kanter FEC, Veldhuis T, Bais B, Schildmeijer R, Ruijter JM, Oostra R-J, Christoffels VM, Moorman AFM. An interactive three-dimensional digital atlas and quantitative database of human development. *Science* 354: aag0053, 2016. doi:10.1126/science.aag0053.

40. Desai TJ, Brownfield DG, Krasnow MA. Alveolar progenitor and stem cells in lung development, renewal and cancer. *Nature* 507: 190–194, 2014. doi:10.1038/nature12930.

41. Domyan ET, Sun X. Patterning and plasticity in development of the respiratory lineage. *Dev Dyn* 240: 477–485, 2011. doi:10.1002/dvdy.22504.

42. Dye BR, Dedhia PH, Miller AJ, Nagy MS, White ES, Shea LD, Spence JR. A bioengineered niche promotes in vivo engraftment and maturation of pluripotent stem cell derived human lung organoids. *eLife* 5: e19732, 2016. doi:10.7554/eLife.19732.

43. Dye BR, Hill DR, Ferguson MA, Tsai Y-H, Nagy MS, Dyal R, Wells JM, Mayhew CN, Nattiv R, Klein OD, White ES, Deutsch GH, Spence JR. In vitro generation of human pluripotent stem cell derived lung organoids. *eLife* 4: e05098, 2015. doi:10.7554/eLife.05098.

44. Dye BR, Miller AJ, Spence JR. How to Grow a Lung: Applying Principles of Developmental Biology to Generate Lung Lineages from Human Pluripotent Stem Cells. *Curr Pathobiol Rep* 4: 47–57, 2016. doi:10.1007/s40139-016-0102-x.

45. Ebina M, Takahashi T, Chiba T, Motomiya M. Cellular hypertrophy and hyperplasia of airway smooth muscles underlying bronchial asthma. A 3-D morphometric study. *Am Rev Respir Dis* 148: 720–726, 1993. doi:10.1164/ajrccm/148.3.720.

46. Finkbeiner SR, Freeman JJ, Wieck MM, El-Nachef W, Altheim CH, Tsai Y-H, Huang S, Dyal R, White ES, Grikscheit TC, Teitelbaum DH, Spence JR. Generation of tissue-engineered small intestine using embryonic stem cell-derived human intestinal organoids. *Biol Open* 4: 1462–1472, 2015. doi:10.1242/bio.013235.

47. Finkbeiner SR, Hill DR, Altheim CH, Dedhia PH, Taylor MJ, Tsai Y-H, Chin AM, Mahe MM, Watson CL, Freeman JJ, Nattiv R, Thomson M, Klein OD, Shroyer NF, Helmrath MA, Teitelbaum DH, Dempsey PJ, Spence JR. Transcriptome-wide analysis reveals hallmarks of human intestine development and maturation in vitro and in vivo. *Stem Cell Reports* 4: 1140–1155, 2015. doi:10.1016/j.stemcr.2015.04.010.

48. Firth AL, Menon T, Parker GS, Qualls SJ, Lewis BM, Ke E, Dargitz CT, Wright R, Khanna A, Gage FH, Verma IM. Functional gene correction for cystic fibrosis in lung epithelial cells generated from patient iPSCs. *Cell Reports* 12: 1385–1390, 2015. doi:10.1016/j.celrep.2015.07.062.

49. Franzdóttir SR, Axelsson IT, Arason AJ, Baldursson O, Gudjonsson T, Magnusson MK. Airway branching morphogenesis in three dimensional culture. *Respir Res* 11: 162, 2010. doi:10.1186/1465-9921-11-162.

50. Freem LJ, Escot S, Tannahill D, Druckenbrod NR, Thapar N, Burns AJ. The intrinsic innervation of the lung is derived from neural crest cells as shown by optical projection tomography in Wnt1-Cre;YFP reporter mice. *J Anat* 217: 651–664, 2010. doi:10.1111/j.1469-7580.2010.01295.x.

51. Fulcher ML, Gabriel SE, Olsen JC, Tatreau JR, Gentzsch M, Livanos E, Saavedra MT, Salmon P, Randell SH. Novel human bronchial epithelial cell lines for cystic fibrosis research. *Am J Physiol Lung Cell Mol Physiol* 296: L82–L91, 2009. doi:10.1152/ajplung.90314.2008.

52. Fulcher ML, Randell SH. Human nasal and tracheo-bronchial respiratory epithelial cell culture. *Methods Mol Biol* 945: 109–121, 2013. doi:10.1007/978-1-62703-125-7_8.

53. Ghaedi M, Calle EA, Mendez JJ, Gard AL, Balestrini J, Booth A, Bove PF, Gui L, White ES, Niklason LE. Human iPS cell-derived alveolar epithelium repopulates lung extracellular matrix. *J Clin Invest* 123: 4950–4962, 2013. doi:10.1172/JCI68793.

54. Ghaedi M, Mendez JJ, Bove PF, Sivarapatna A, Raredon MSB, Niklason LE. Alveolar epithelial differentiation of human induced pluripotent stem cells in a rotating bioreactor. *Biomaterials* 35: 699–710, 2014. doi:10.1016/j.biomaterials.2013.10.018.

55. Ghaedi M, Niklason LE, Williams J. Development of lung epithelium from induced pluripotent stem cells. *Curr Transplant Rep* 2: 81–89, 2015. doi:10.1007/s40472-014-0039-0.

56. Gilpin SE, Charest JM, Ren X, Ott HC. Bioengineering lungs for transplantation. *Thorac Surg Clin* 26: 163–171, 2016. doi:10.1016/j.thorsurg.2015.12.004.

57. Gilpin SE, Ott HC. Using nature's platform to engineer bio-artificial lungs. *Ann Am Thorac Soc* 12, *Suppl* 1: S45–S49, 2015. doi:10.1513/AnnalsATS.201408-366MG.

Downloaded from http://physiologyonline.physiology.org/ by 10.220.33.6 on April 15, 2017

58. Gilpin SE, Ren X, Okamoto T, Guyette JP, Mou H, Rajagopal J, Mathisen DJ, Vacanti JP, Ott HC. Enhanced lung epithelial specification of human induced pluripotent stem cells on decellularized lung matrix. *Ann Thorac Surg* 98: 1721–1729, 2014.

59. Gjorevski N, Sachs N, Manfrin A, Giger S, Bragina ME, Ordóñez-Morán P, Clevers H, Lutolf MP. Designer matrices for intestinal stem cell and organoid culture. *Nature* 539: 560–564, 2016. doi:10.1038/nature20168.

60. Glasser SW, Hagood JS, Wong S, Taype CA, Madala SK, Hardie WD. Mechanisms of lung fibrosis resolution. *Am J Pathol* 186: 1066–1077, 2016. doi:10.1016/j.ajpath.2016.01.018.

61. Gonzalez RF, Allen L, Gonzales L, Ballard PL, Dobbs LG. HTII-280, a biomarker specific to the apical plasma membrane of human lung alveolar type II cells. *J Histochem Cytochem* 58: 891–901, 2010. doi:10.1369/jhc.2010.956433.

62. Gotoh S, Ito I, Nagasaki T, Yamamoto Y, Konishi S, Korogi Y, Matsumoto H, Muro S, Hirai T, Funato M, Mae S, Toyoda T, Sato-Otsubo A, Ogawa S, Osafune K, Mishima M. Generation of alveolar epithelial spheroids via isolated progenitor cells from human pluripotent stem cells. *Stem Cell Reports* 3: 394–403, 2014. doi:10.1016/j.stemcr.2014.07.005.

63. Green J, Endale M, Auer H, Perl AK. Diversity of interstitial lung fibroblasts is regulated by platelet-derived growth factor receptor α kinase activity. *Am J Respir Cell Mol Biol* 54: 532–545, 2016. doi:10.1165/rcmb.2015-0095OC.

64. Green MD, Chen A, Nostro M-C, d'Souza SL, Schaniel C, Lemischka IR, Gouon-Evans V, Keller G, Snoeck H-W. Generation of anterior foregut endoderm from human embryonic and induced pluripotent stem cells. *Nat Biotechnol* 29: 267–272, 2011. doi:10.1038/nbt.1788.

65. Gruenert DC, Willems M, Cassiman JJ, Frizzell RA. Established cell lines used in cystic fibrosis research. *J Cyst Fibros* 3, *Suppl* 2: 191–196, 2004. doi:10.1016/j.jcf.2004.05.040.

66. Gu X, Karp PH, Brody SL, Pierce RA, Welsh MJ, Holtzman MJ, Ben-Shahar Y. Chemosensory functions for pulmonary neuroendocrine cells. *Am J Respir Cell Mol Biol* 50: 637–646, 2014. doi:10.1165/rcmb.2013-0199OC.

67. Guyette JP, Gilpin SE, Charest JM, Tapias LF, Ren X, Ott HC. Perfusion decellularization of whole organs. *Nat Protoc* 9: 1451–1468, 2014. doi:10.1038/nprot.2014.097.

68. Hackett T-L, Shaheen F, Johnson A, Wadsworth S, Pechkovsky DV, Jacoby DB, Kicic A, Stick SM, Knight DA. Characterization of side population cells from human airway epithelium. *Stem Cells* 26: 2576–2585, 2008. doi:10.1634/stemcells.2008-0171.

69. Han RNN, Babaei S, Robb M, Lee T, Ridsdale R, Ackerley C, Post M, Stewart DJ. Defective lung vascular development and fatal respiratory distress in endothelial NO synthase-deficient mice: a model of alveolar capillary dysplasia? *Circ Res* 94: 1115–1123, 2004. doi:10.1161/01.RES.0000125624.85852.1E.

70. Hannan NRF, Fordham RP, Syed YA, Moignard V, Berry A, Bautista R, Hanley NA, Jensen KB, Vallier L. Generation of multipotent foregut stem cells from human pluripotent stem cells. *Stem Cell Reports* 1: 293–306, 2013. doi:10.1016/j.stemcr.2013.09.003.

71. Hannan NRF, Sampaziotis F, Segeritz C-P, Hanley NA, Vallier L. Generation of distal airway epithelium from multipotent human foregut stem cells. *Stem Cells Dev* 24: 1680–1690, 2015. doi:10.1089/scd.2014.0512.

72. Harunaga JS, Doyle AD, Yamada KM. Local and global dynamics of the basement membrane during branching morphogenesis require protease activity and actomyosin contractility. *Dev Biol* 394: 197–205, 2014. doi:10.1016/j.ydbio.2014.08.014.

73. Haworth SG, Hislop AA. Lung development-the effects of chronic hypoxia. *Semin Neonatol* 8: 1–8, 2003. doi:10.1016/S1084-2756(02)00195-1.

74. Hegab AE, Arai D, Gao J, Kuroda A, Yasuda H, Ishii M, Naoki K, Soejima K, Betsuyaku T. Mimicking the niche of lung epithelial stem cells and characterization of several effectors of their in vitro behavior. *Stem Cell Res (Amst)* 15: 109–121, 2015. doi:10.1016/j.scr.2015.05.005.

75. Hermanns MI, Unger RE, Kehe K, Peters K, Kirkpatrick CJ. Lung epithelial cell lines in coculture with human pulmonary microvascular endothelial cells: development of an alveolo-capillary barrier in vitro. *Lab Invest* 84: 736–752, 2004. doi:10.1038/labinvest.3700081.

76. Herriges M, Morrisey EE. Lung development: orchestrating the generation and regeneration of a complex organ. *Development* 141: 502–513, 2014. doi:10.1242/dev.098186.

77. Hild M, Jaffe AB. Production of 3-D airway organoids from primary human airway basal cells and their use in high-throughput screening. *Curr Protoc Stem Cell Biol* 37: IE.9.1.9.15502–IE, 2016. doi:10.1002/cpsc.1.

78. Hoegger MJ, Fischer AJ, McMenimen JD, Ostedgaard LS, Tucker AJ, Awadalla MA, Moninger TO, Michalski AS, Hoffman EA, Zabner J, Stoltz DA, Welsh MJ. Impaired mucus detachment disrupts mucociliary transport in a piglet model of cystic fibrosis. *Science* 345: 818–822, 2014. doi:10.1126/science.1255825.

79. Holden VI, Breen P, Houle S, Dozois CM, Bachman MA. Klebsiella pneumoniae Siderophores induce inflammation, bacterial dissemination, and HIF-1α stabilization during pneumonia. *MBio* 7: e01397-16, 2016. doi:10.1128/mBio.01397-16.

80. Hong KU, Reynolds SD, Watkins S, Fuchs E, Stripp BR. Basal cells are a multipotent progenitor capable of renewing the bronchial epithelium. *Am J Pathol* 164: 577–588, 2004. doi:10.1016/S0002-9440(10)63147-1.

81. Hoo AF, Henschen M, Dezateux C, Costeloe K, Stocks J. Respiratory function among preterm infants whose mothers smoked during pregnancy. *Am J Respir Crit Care Med* 158: 700–705, 1998. doi:10.1164/ajrccm.158.3.9711057.

82. Huang SXL, Islam MN, O'Neill J, Hu Z, Yang Y-G, Chen Y-W, Mumau M, Green MD, Vunjak-Novakovic G, Bhattacharya J, Snoeck H-W. Efficient generation of lung and airway epithelial cells from human pluripotent stem cells. *Nat Biotechnol* 32: 84–91, 2014. doi:10.1038/nbt.2754.

83. Huh D, Leslie DC, Matthews BD, Fraser JP, Jurek S, Hamilton GA, Thorneloe KS, McAlexander MA, Ingber DE. A human disease model of drug toxicity-induced pulmonary edema in a lung-on-a-chip microdevice. *Sci Transl Med* 4: 159ra147, 2012. doi:10.1126/scitranslmed.3004249.

84. Huh D, Matthews BD, Mammoto A, Montoya-Zavala M, Hsin HY, Ingber DE. Reconstituting organ-level lung functions on a chip. *Science* 328: 1662–1668, 2010. doi:10.1126/science.1188302.

85. Jain R, Barkauskas CE, Takeda N, Bowie EJ, Aghajanian H, Wang Q, Padmanabhan A, Manderfield LJ, Gupta M, Li D, Li L, Trivedi CM, Hogan BLM, Epstein JA. Plasticity of Hopx(+) type I alveolar cells to regenerate type II cells in the lung. *Nat Commun* 6: 6727, 2015. doi:10.1038/ncomms7727.

86. Kim CFB, Jackson EL, Woolfenden AE, Lawrence S, Babar I, Vogel S, Crowley D, Bronson RT, Jacks T. Identification of bronchioalveolar stem cells in normal lung and lung cancer. *Cell* 121: 823–835, 2005. doi:10.1016/j.cell.2005.03.032.

87. Kim HY, Nelson CM. Extracellular matrix and cytoskeletal dynamics during branching morphogenesis. *Organogenesis* 8: 56–64, 2012.

88. Kim HY, Pang M-F, Varner VD, Kojima L, Miller E, Radisky DC, Nelson CM. Localized smooth muscle differentiation is essential for epithelial bifurcation during branching morphogenesis of the mammalian lung. *Dev Cell* 34: 719–726, 2015. doi:10.1016/j.devcel.2015.08.012.

89. Knowles MR, Boucher RC. Mucus clearance as a primary innate defense mechanism for mammalian airways. *J Clin Invest* 109: 571–577, 2002. doi:10.1172/JCI0215217.

90. Konishi S, Gotoh S, Tateishi K, Yamamoto Y, Korogi Y, Nagasaki T, Matsumoto H, Muro S, Hirai T, Ito I, Tsukita S, Mishima M. Directed induction of functional multi-ciliated cells in proximal airway epithelial spheroids from human pluripotent stem cells. *Stem Cell Reports* 6: 18–25, 2016. doi:10.1016/j.stemcr.2015.11.010.

91. Kotton DN, Morrisey EE. Lung regeneration: mechanisms, applications and emerging stem cell populations. *Nat Med* 20: 822–832, 2014. doi:10.1038/nm.3642.

92. Kroon E, Martinson LA, Kadoya K, Bang AG, Kelly OG, Eliazer S, Young H, Richardson M, Smart NG, Cunningham J, Agulnick AD, D'Amour KA, Carpenter MK, Baetge EE. Pancreatic endoderm derived from human embryonic stem cells generates glucose-responsive insulin-secreting cells in vivo. *Nat Biotechnol* 26: 443–452, 2008. doi:10.1038/nbt1393.

93. Lee J-H, Bhang DH, Beede A, Huang TL, Stripp BR, Bloch KD, Wagers AJ, Tseng Y-H, Ryeom S, Kim CF. Lung stem cell differentiation in mice directed by endothelial cells via a BMP4-NFATc1-thrombospondin-1 axis. *Cell* 156: 440–455, 2014. doi:10.1016/j.cell.2013.12.039.

94. LeSimple P, Liao J, Robert R, Gruenert DC, Hanrahan JW. Cystic fibrosis transmembrane conductance regulator trafficking modulates the barrier function of airway epithelial cell monolayers. *J Physiol* 588: 1195–1209, 2010. doi:10.1113/jphysiol.2009.182246.

95. Liu X, Ory V, Chapman S, Yuan H, Albanese C, Kallakury B, Timofeeva OA, Nealon C, Dakic A, Simic V, Haddad BR, Rhim JS, Dritschilo A, Riegel A, McBride A, Schlegel R. ROCK inhibitor and feeder cells induce the conditional reprogramming of epithelial cells. *Am J Pathol* 180: 599–607, 2012. doi:10.1016/j.ajpath.2011.10.036.

96. Longmire TA, Ikonomou L, Hawkins F, Christodoulou C, Cao Y, Jean JC, Kwok LW, Mou H, Rajagopal J, Shen SS, Dowton AA, Serra M, Weiss DJ, Green MD, Snoeck H-W, Ramirez MI, Kotton DN. Efficient derivation of purified lung and thyroid progenitors from embryonic stem cells. *Cell Stem Cell* 10: 398–411, 2012. doi:10.1016/j.stem.2012.01.019.

97. Mak IW, Evaniew N, Ghert M. Lost in translation: animal models and clinical trials in cancer treatment. *Am J Transl Res* 6: 114–118, 2014.

98. Massaro D, Massaro GD. Invited Review: pulmonary alveoli: formation, the "call for oxygen," and other regulators. *Am J Physiol Lung Cell Mol Physiol* 282: L345–L358, 2002. doi:10.1152/ajplung.00374.2001.

99. Matute-Bello G, Frevert CW, Martin TR. Animal models of acute lung injury. *Am J Physiol Lung Cell Mol Physiol* 295: L379–L399, 2008. doi:10.1152/ajplung.00010.2008.

100. McCracken KW, Catá EM, Crawford CM, Sinagoga KL, Schumacher M, Rockich BE, Tsai Y-H, Mayhew CN, Spence JR, Zavros Y, Wells JM. Modelling human development and disease in pluripotent stem-cell-derived gastric organoids. *Nature* 516: 400–404, 2014. doi:10.1038/nature13863.

Downloaded from http://physiologyonline.physiology.org/ by 10.220.33.6 on April 15, 2017

101. McCracken KW, Howell JC, Wells JM, Spence JR. Generating human intestinal tissue from pluripotent stem cells in vitro. *Nat Protoc* 6: 1920–1928, 2011. doi:10.1038/nprot.2011.410.

102. McCulley D, Wienhold M, Sun X. The pulmonary mesenchyme directs lung development. *Curr Opin Genet Dev* 32: 98–105, 2015. doi:10.1016/j.gde.2015.01.011.

103. McGowan SE. Extracellular matrix and the regulation of lung development and repair. *FASEB J* 6: 2895–2904, 1992.

104. McQualter JL, Bertoncello I. Clonal culture of adult mouse lung epithelial stem/progenitor cells. *Methods Mol Biol* 1235: 231–241, 2015. doi:10.1007/978-1-4939-1785-3_17.

105. Mercer RR, Russell ML, Roggli VL, Crapo JD. Cell number and distribution in human and rat airways. *Am J Respir Cell Mol Biol* 10: 613–624, 1994. doi:10.1165/ajrcmb.10.6.8003339.

106. Metzger RJ, Klein OD, Martin GR, Krasnow MA. The branching programme of mouse lung development. *Nature* 453: 745–750, 2008. doi:10.1038/nature07005.

107. Molina SA, Stauffer B, Moriarty HK, Kim AH, McCarty NA, Koval M. Junctional abnormalities in human airway epithelial cells expressing F508del CFTR. *Am J Physiol Lung Cell Mol Physiol* 309: L475–L487, 2015. doi:10.1152/ajplung.00060.2015.

108. Mondrinos MJ, Koutzaki S, Lelkes PI, Finck CM. A tissue-engineered model of fetal distal lung tissue. *Am J Physiol Lung Cell Mol Physiol* 293: L639–L650, 2007. doi:10.1152/ajplung.00403.2006.

109. Moore BB, Hogaboam CM. Murine models of pulmonary fibrosis. *Am J Physiol Lung Cell Mol Physiol* 294: L152–L160, 2008. doi:10.1152/ajplung.00313.2007.

110. Morris GE, Bridge JC, Brace LA, Knox AJ, Aylott JW, Brightling CE, Ghaemmaghami AM, Rose FRAJ. A novel electrospun biphasic scaffold provides optimal three-dimensional topography for in vitro co-culture of airway epithelial and fibroblast cells. *Biofabrication* 6: 035014, 2014. doi:10.1088/1758-5082/6/3/035014.

111. Morrisey EE, Hogan BLM. Preparing for the first breath: genetic and cellular mechanisms in lung development. *Dev Cell* 18: 8–23, 2010. doi:10.1016/j.devcel.2009.12.010.

112. Mou H, Vinarsky V, Tata PR, Brazauskas K, Choi SH, Crooke AK, Zhang B, Solomon GM, Turner B, Bihler H, Harrington J, Lapey A, Channick C, Keyes C, Freund A, Artandi S, Mense M, Rowe S, Engelhardt JF, Hsu Y-C, Rajagopal J. Dual SMAD signaling inhibition enables long-term expansion of diverse epithelial basal cells. *Cell Stem Cell* 19: 217–231, 2016. doi:10.1016/j.stem.2016.05.012.

113. Mou H, Zhao R, Sherwood R, Ahfeldt T, Lapey A, Wain J, Sicilian L, Izvolsky K, Musunuru K, Cowan C, Rajagopal J. Generation of multipotent lung and airway progenitors from mouse ESCs and patient-specific cystic fibrosis iPSCs. *Cell Stem Cell* 10: 385–397, 2012. doi:10.1016/j.stem.2012.01.018.

114. Myllärniemi M, Kaarteenaho R. Pharmacological treatment of idiopathic pulmonary fibrosis - preclinical and clinical studies of pirfenidone, nintedanib, and N-acetylcysteine. *Eur Clin Respir J* 2: 26385, 2015. doi:10.3402/ecrj.v2.26385.

115. Enemchukwu NO, Cruz-Acuña R, Bongiorno T, Johnson CT, García JR, Sulchek T, García AJ. Synthetic matrices reveal contributions of ECM biophysical and biochemical properties to epithelial morphogenesis. *J Cell Biol* 212: 113–124, 2016. doi:10.1083/jcb.201506055.

116. Neuberger T, Burton B, Clark H, Van Goor F. Use of primary cultures of human bronchial epithelial cells isolated from cystic fibrosis patients for the pre-clinical testing of CFTR modulators. *Methods Mol Biol* 741: 39–54, 2011. doi:10.1007/978-1-61779-117-8_4.

117. O'Leary C, Cavanagh B, Unger RE, Kirkpatrick CJ, O'Dea S, O'Brien FJ, Cryan S-A. The development of a tissue-engineered tracheobronchial epithelial model using a bilayered collagen-hyaluronate scaffold. *Biomaterials* 85: 111–127, 2016. doi:10.1016/j.biomaterials.2016.01.065.

118. Ott HC, Clippinger B, Conrad C, Schuetz C, Pomerantseva I, Ikonomou L, Kotton D, Vacanti JP. Regeneration and orthotopic transplantation of a bioartificial lung. *Nat Med* 16: 927–933, 2010. doi:10.1038/nm.2193.

119. Pageau SC, Sazonova OV, Wong JY, Soto AM, Sonnenschein C. The effect of stromal components on the modulation of the phenotype of human bronchial epithelial cells in 3D culture. *Biomaterials* 32: 7169–7180, 2011. doi:10.1016/j.biomaterials.2011.06.017.

120. Pardo-Saganta A, Law BM, Gonzalez-Celeiro M, Vinarsky V, Rajagopal J. Ciliated cells of pseudostratified airway epithelium do not become mucous cells after ovalbumin challenge. *Am J Respir Cell Mol Biol* 48: 364–373, 2013. doi:10.1165/rcmb.2012-0146OC.

121. Pardo-Saganta A, Law BM, Tata PR, Villoria J, Saez B, Mou H, Zhao R, Rajagopal J. Injury induces direct lineage segregation of functionally distinct airway basal stem/progenitor cell subpopulations. *Cell Stem Cell* 16: 184–197, 2015. doi:10.1016/j.stem.2015.01.002.

122. Patel VN, Pineda DL, Hoffman MP. The function of heparan sulfate during branching morphogenesis. *Matrix Biol* 57-58: 311–323, 2017. doi:10.1016/j.matbio.2016.09.004.

123. Peng T, Frank DB, Kadzik RS, Morley MP, Rathi KS, Wang T, Zhou S, Cheng L, Lu MM, Morrisey EE. Hedgehog actively maintains adult lung quiescence and regulates repair and regeneration. *Nature* 526: 578–582, 2015. doi:10.1038/nature14984.

124. Perrin DG, McDonald TJ, Cutz E. Hyperplasia of bombesin-immunoreactive pulmonary neuroendocrine cells and neuroepithelial bodies in sudden infant death syndrome. *Pediatr Pathol* 11: 431–447, 1991. doi:10.3109/15513819109064779.

125. Perrin S. Preclinical research: make mouse studies work. *Nature* 507: 423–425, 2014. doi:10.1038/507423a.

126. Petersen TH, Calle EA, Zhao L, Lee EJ, Gui L, Raredon MB, Gavrilov K, Yi T, Zhuang ZW, Breuer C, Herzog E, Niklason LE. Tissue-engineered lungs for in vivo implantation. *Science* 329: 538–541, 2010. doi:10.1126/science.1189345.

127. Pifferi M, Montemurro F, Cangiotti AM, Ragazzo V, Di Cicco M, Vinci B, Vozzi G, Macchia P, Boner AL. Simplified cell culture method for the diagnosis of atypical primary ciliary dyskinesia. *Thorax* 64: 1077–1081, 2009. doi:10.1136/thx.2008.110940.

128. Pizurki L, Morris MA, Chanson M, Solomon M, Pavirani A, Bouchardy I, Suter S. Cystic fibrosis transmembrane conductance regulator does not affect neutrophil migration across cystic fibrosis airway epithelial monolayers. *Am J Pathol* 156: 1407–1416, 2000. doi:10.1016/S0002-9440(10)65009-2.

129. Prakash YS, Tschumperlin DJ, Stenmark KR. Coming to terms with tissue engineering and regenerative medicine in the lung. *Am J Physiol Lung Cell Mol Physiol* 309: L625–L638, 2015. doi:10.1152/ajplung.00204.2015.

130. Rackley CR, Stripp BR. Building and maintaining the epithelium of the lung. *J Clin Invest* 122: 2724–2730, 2012. doi:10.1172/JCI60519.

131. Randell SH, Fulcher ML, O'Neal W, Olsen JC. Primary epithelial cell models for cystic fibrosis research. *Methods Mol Biol* 742: 285–310, 2011. doi:10.1007/978-1-61779-120-8_18.

132. Rawlins EL, Clark CP, Xue Y, Hogan BLM. The Id2+ distal tip lung epithelium contains individual multipotent embryonic progenitor cells. *Development* 136: 3741–3745, 2009. doi:10.1242/dev.037317.

133. Rawlins EL, Okubo T, Xue Y, Brass DM, Auten RL, Hasegawa H, Wang F, Hogan BLM. The role of Scgb1a1+ Clara cells in the long-term maintenance and repair of lung airway, but not alveolar, epithelium. *Cell Stem Cell* 4: 525–534, 2009. doi:10.1016/j.stem.2009.04.002.

134. Rawlins EL, Ostrowski LE, Randell SH, Hogan BLM. Lung development and repair: contribution of the ciliated lineage. *Proc Natl Acad Sci USA* 104: 410–417, 2007. doi:10.1073/pnas.0610770104.

136. Rawlins EL. The building blocks of mammalian lung development. *Dev Dyn* 240: 463–476, 2011. doi:10.1002/dvdy.22482.

137. Ren H, Birch NP, Suresh V. An optimised human cell culture model for alveolar epithelial transport. *PLoS One* 11: e0165225, 2016. doi:10.1371/journal.pone.0165225.

138. Ren X, Moser PT, Gilpin SE, Okamoto T, Wu T, Tapias LF, Mercier FE, Xiong L, Ghawi R, Scadden DT, Mathisen DJ, Ott HC. Engineering pulmonary vasculature in decellularized rat and human lungs. *Nat Biotechnol* 33: 1097–1102, 2015. doi:10.1038/nbt.3354.

139. Rock JR, Gao X, Xue Y, Randell SH, Kong Y-Y, Hogan BLM. Notch-dependent differentiation of adult airway basal stem cells. *Cell Stem Cell* 8: 639–648, 2011. doi:10.1016/j.stem.2011.04.003.

140. Rock JR, Onaitis MW, Rawlins EL, Lu Y, Clark CP, Xue Y, Randell SH, Hogan BLM. Basal cells as stem cells of the mouse trachea and human airway epithelium. *Proc Natl Acad Sci USA* 106: 12771–12775, 2009. doi:10.1073/pnas.0906850106.

141. Rockich BE, Hrycaj SM, Shih HP, Nagy MS, Ferguson MAH, Kopp JL, Sander M, Wellik DM, Spence JR. Sox9 plays multiple roles in the lung epithelium during branching morphogenesis. *Proc Natl Acad Sci USA* 110: E4456–E4464, 2013. doi:10.1073/pnas.1311847110.

142. Ruggeri BA, Camp F, Miknyoczki S. Animal models of disease: pre-clinical animal models of cancer and their applications and utility in drug discovery. *Biochem Pharmacol* 87: 150–161, 2014. doi:10.1016/j.bcp.2013.06.020.

143. Rutland J, Cole PJ. Non-invasive sampling of nasal cilia for measurement of beat frequency and study of ultrastructure. *Lancet* 2: 564–565, 1980. doi:10.1016/S0140-6736(80)91995-9.

144. Sakao S, Tatsumi K, Voelkel NF. Endothelial cells and pulmonary arterial hypertension: apoptosis, proliferation, interaction and transdifferentiation. *Respir Res* 10: 95, 2009. doi:10.1186/1465-9921-10-95.

145. Schamberger AC, Staab-Weijnitz CA, Mise-Racek N, Eickelberg O. Cigarette smoke alters primary human bronchial epithelial cell differentiation at the air-liquid interface. *Sci Rep* 5: 8163, 2015. doi:10.1038/srep08163.

146. Schwank G, Koo B-K, Sasselli V, Dekkers JF, Heo I, Demircan T, Sasaki N, Boymans S, Cuppen E, van der Ent CK, Nieuwenhuis EES, Beekman JM, Clevers H. Functional repair of CFTR by CRISPR/Cas9 in intestinal stem cell organoids of cystic fibrosis patients. *Cell Stem Cell* 13: 653–658, 2013. doi:10.1016/j.stem.2013.11.002.

147. Sekhon HS, Jia Y, Raab R, Kuryatov A, Pankow JF, Whitsett JA, Lindstrom J, Spindel ER. Prenatal nicotine increases pulmonary alpha7 nicotinic receptor expression and alters fetal lung development in monkeys. *J Clin Invest* 103: 637–647, 1999. doi:10.1172/JCI5232.

Downloaded from http://physiologyonline.physiology.org/ by 10.220.33.6 on April 15, 2017

148. Sekine K, Ohuchi H, Fujiwara M, Yamasaki M, Yoshizawa T, Sato T, Yagishita N, Matsui D, Koga Y, Itoh N, Kato S. Fgf10 is essential for limb and lung formation. *Nat Genet* 21: 138–141, 1999. doi:10.1038/5096.

149. Shah VS, Meyerholz DK, Tang XX, Reznikov L, Abou Alaiwa M, Ernst SE, Karp PH, Wohlford-Lenane CL, Heilmann KP, Leidinger MR, Allen PD, Zabner J, McCray PB Jr, Ostedgaard LS, Stoltz DA, Randak CO, Welsh MJ. Airway acidification initiates host defense abnormalities in cystic fibrosis mice. *Science* 351: 503–507, 2016. doi:10.1126/science.aad5589.

150. Snider GL. Collagen concentration and rates of synthesis in idiopathic pulmonary fibrosis. *Am Rev Respir Dis* 124: 341–342, 1981.

151. Snoeck H-W. Modeling human lung development and disease using pluripotent stem cells. *Development* 142: 13–16, 2015. doi:10.1242/dev.115469.

152. Spence JR, Mayhew CN, Rankin SA, Kuhar MF, Vallance JE, Tolle K, Hoskins EE, Kalinichenko VV, Wells SI, Zorn AM, Shroyer NF, Wells JM. Directed differentiation of human pluripotent stem cells into intestinal tissue in vitro. *Nature* 470: 105–109, 2011. doi:10.1038/nature09691.

153. Stabler CT, Caires LC Jr, Mondrinos MJ, Marcinkiewicz C, Lazarovici P, Wolfson MR, Lelkes PI. Enhanced re-endothelialization of decellularized rat lungs. *Tissue Eng Part C Methods* 22: 439–450, 2016. doi:10.1089/ten.tec.2016.0012.

154. Tadokoro T, Gao X, Hong CC, Hotten D, Hogan BLM. BMP signaling and cellular dynamics during regeneration of airway epithelium from basal progenitors. *Development* 143: 764–773, 2016. doi:10.1242/dev.126656.

155. Takasato M, Er PX, Chiu HS, Maier B, Baillie GJ, Ferguson C, Parton RG, Wolvetang EJ, Roost MS, Chuva de Sousa Lopes SM, Little MH. Kidney organoids from human iPS cells contain multiple lineages and model human nephrogenesis. *Nature* 526: 564–568, 2015. doi:10.1038/nature15695.

156. Tan Q, Choi KM, Sicard D, Tschumperlin DJ. Human airway organoid engineering as a step toward lung regeneration and disease modeling. *Biomaterials* 113: 118–132, 2017. doi:10.1016/j.biomaterials.2016.10.046.

157. Tata PR, Mou H, Pardo-Saganta A, Zhao R, Prabhu M, Law BM, Vinarsky V, Cho JL, Breton S, Sahay A, Medoff BD, Rajagopal J. Dedifferentiation of committed epithelial cells into stem cells in vivo. *Nature* 503: 218–223, 2013. doi:10.1038/nature12777.

158. Thébaud B, Ladha F, Michelakis ED, Sawicka M, Thurston G, Eaton F, Hashimoto K, Harry G, Haromy A, Korbutt G, Archer SL. Vascular endothelial growth factor gene therapy increases survival, promotes lung angiogenesis, and prevents alveolar damage in hyperoxia-induced lung injury: evidence that angiogenesis participates in alveolarization. *Circulation* 112: 2477–2486, 2005. doi:10.1161/CIRCULATIONAHA.105.541524.

159. Tilley AE, Walters MS, Shaykhiev R, Crystal RG. Cilia dysfunction in lung disease. *Annu Rev Physiol* 77: 379–406, 2015. doi:10.1146/annurev-physiol-021014-071931.

160. Toskala E, Haataja J, Shirasaki H, Rautiainen M. Culture of cells harvested with nasal brushing: a method for evaluating ciliary function. *Rhinology* 43: 121–124, 2005.

161. Tovar JA. Congenital diaphragmatic hernia. *Orphanet J Rare Dis* 7: 1, 2012. doi:10.1186/1750-1172-7-1.

162. Trecartin A, Danopoulos S, Spurrier R, Knaneh-Monem H, Hiatt M, Driscoll B, Hochstim C, Al-Alam D, Grikscheit TC. Establishing proximal and distal regional identities in murine and human tissue-engineered lung and trachea. *Tissue Eng Part C Methods* 22: 1049–1057, 2016. doi:10.1089/ten.tec.2016.0261.

163. Truog WE, Xu D, Ekekezie II, Mabry S, Rezaiekhaligh M, Svojanovsky S, Soares MJ. Chronic hypoxia and rat lung development: analysis by morphometry and directed microarray. *Pediatr Res* 64: 56–62, 2008. doi:10.1203/PDR.0b013e31817289f2.

164. Van Goor F, Hadida S, Grootenhuis PDJ, Burton B, Cao D, Neuberger T, Turnbull A, Singh A, Joubran J, Hazlewood A, Zhou J, McCartney J, Arumugam V, Decker C, Yang J, Young C, Olson ER, Wine JJ, Frizzell RA, Ashlock M, Negulescu P. Rescue of CF airway epithelial cell function in vitro by a CFTR potentiator, VX-770. *Proc Natl Acad Sci USA* 106: 18825–18830, 2009. doi:10.1073/pnas.0904709106.

165. Varner VD, Gleghorn JP, Miller E, Radisky DC, Nelson CM. Mechanically patterning the embryonic airway epithelium. *Proc Natl Acad Sci USA* 112: 9230–9235, 2015. doi:10.1073/pnas.1504102112.

166. Varner VD, Nelson CM. Cellular and physical mechanisms of branching morphogenesis. *Development* 141: 2750–2759, 2014. doi:10.1242/dev.104794.

167. Vaughan AE, Brumwell AN, Xi Y, Gotts JE, Brownfield DG, Treutlein B, Tan K, Tan V, Liu FC, Looney MR, Matthay MA, Rock JR, Chapman HA. Lineage-negative progenitors mobilize to regenerate lung epithelium after major injury. *Nature* 517: 621–625, 2015. doi:10.1038/nature14112.

168. Vorotnikova E, McIntosh D, Dewilde A, Zhang J, Reing JE, Zhang L, Cordero K, Bedelbaeva K, Gourevitch D, Heber-Katz E, Badylak SF, Braunhut SJ. Extracellular matrix-derived products modulate endothelial and progenitor cell migration and proliferation in vitro and stimulate regenerative healing in vivo. *Matrix Biol* 29: 690–700, 2010. doi:10.1016/j.matbio.2010.08.007.

169. Wagner DE, Bonenfant NR, Parsons CS, Sokocevic D, Brooks EM, Borg ZD, Lathrop MJ, Wallis JD, Daly AB, Lam YW, Deng B, DeSarno MJ, Ashikaga T, Loi R, Weiss DJ. Comparative decellularization and recellularization of normal versus emphysematous human lungs. *Biomaterials* 35: 3281–3297, 2014. doi:10.1016/j.biomaterials.2013.12.103.

170. Watson CL, Mahe MM, Múnera J, Howell JC, Sundaram N, Poling HM, Schweitzer JI, Vallance JE, Mayhew CN, Sun Y, Grabowski G, Finkbeiner SR, Spence JR, Shroyer NF, Wells JM, Helmrath MA. An in vivo model of human small intestine using pluripotent stem cells. *Nat Med* 20: 1310–1314, 2014. doi:10.1038/nm.3737.

171. Weiser N, Molenda N, Urbanova K, Bähler M, Pieper U, Oberleithner H, Schillers H. Paracellular permeability of bronchial epithelium is controlled by CFTR. *Cell Physiol Biochem* 28: 289–296, 2011. doi:10.1159/000331742.

172. Whitcutt MJ, Adler KB, Wu R. A biphasic chamber system for maintaining polarity of differentiation of cultured respiratory tract epithelial cells. *In Vitro Cell Dev Biol* 24: 420–428, 1988. doi:10.1007/BF02628493.

173. White ES. Lung extracellular matrix and fibroblast function. *Ann Am Thorac Soc* 12, *Suppl* 1: S30–S33, 2015. doi:10.1513/AnnalsATS.201406-240MG.

174. Whitsett JA, Kalinichenko VV. Notch and basal cells take center stage during airway epithelial regeneration. *Cell Stem Cell* 8: 597–598, 2011. doi:10.1016/j.stem.2011.05.008.

175. Wilkinson DC, Alva-Ornelas JA, Sucre JMS, Vijayaraj P, Durra A, Richardson W, Jonas SJ, Paul MK, Karumbayaram S, Dunn B, Gomperts BN. Development of a three-dimensional bioengineering technology to generate lung tissue for personalized disease modeling. *Stem Cells Transl Med.* 6: 622–633, 2017. doi:10.5966/sctm.2016-0192.

176. Willem M, Miosge N, Halfter W, Smyth N, Jannetti I, Burghart E, Timpl R, Mayer U. Specific ablation of the nidogen-binding site in the laminin gamma1 chain interferes with kidney and lung development. *Development* 129: 2711–2722, 2002.

177. Wolff RK. Effects of airborne pollutants on mucociliary clearance. *Environ Health Perspect* 66: 223–237, 1986. doi:10.1289/ehp.8666223.

178. Wong AP, Bear CE, Chin S, Pasceri P, Thompson TO, Huan L-J, Ratjen F, Ellis J, Rossant J. Directed differentiation of human pluripotent stem cells into mature airway epithelia expressing functional CFTR protein. *Nat Biotechnol* 30: 876–882, 2012. doi:10.1038/nbt.2328.

179. Workman MJ, Mahe MM, Trisno S, Poling HM, Watson CL, Sundaram N, Chang C-F, Schiesser J, Aubert P, Stanley EG, Elefanty AG, Miyaoka Y, Mandegar MA, Conklin BR, Neunlist M, Brugmann SA, Helmrath MA, Wells JM. Engineered human pluripotent-stem-cell-derived intestinal tissues with a functional enteric nervous system. *Nat Med* 23: 49–59, 2017. doi:10.1038/nm.4233.

180. Yamaya M, Finkbeiner WE, Chun SY, Widdicombe JH. Differentiated structure and function of cultures from human tracheal epithelium. *Am J Physiol Lung Cell Mol Physiol* 262: L713–L724, 1992.

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